

How we fit λ , γ , and temperature — read this first

For: Charles — learn the MLE thread from scratch
Date: 2026-08-20
Go deeper (derivations): `MLE_basics.md`
Slides: `../re_entry/HER0s_and_PASSes/slides/auto/CHAR_PD21_MLE_fit_AUTO.pptx`
Code: `sports/scripts/pd21_draft_bernoulli_mle.py`

What this document is for

You asked for **instruction**, not a cheat sheet. This page walks through **what we did**, **why**, and **what the numbers mean**, assuming you remember the rest of the pipeline (ASSIGN / SCORE / SELECT, hero, p^* , etc.) but **not** the statistics of maximum likelihood yet. Symbols like λ , t , γ , A , L^C are fine — we use those everywhere. Words like “logit” and “Bernoulli MLE” are **not** assumed; when they appear, they get a one-line meaning, and the full derivation lives in `MLE_basics.md`.

The question we answered

*Given real NCAA player-seasons (drafted or not), **what values of λ and t make our draft-probability story best match what actually happened?***

We held $\gamma = 18$ fixed (same as the generative sim default). We did **not** re-fit p — rosters stay as observed.

Panel (committed run): 2013–2021, POST-QC, min 20 minutes → **38,123** player-seasons, **882** drafted.

Where this sits in the three-step pipeline

You already know the split:

Step	Knob(s)	This fit?
ASSIGN	p (who lands on which roster)	No — p^* from H_{sort} bracket separately
SCORE	λ , γ (ability vs congestion in the ranking formula)	Partially — λ yes; γ fixed at 18
SELECT	t (how sharply probabilities follow the score)	Yes — t from MLE

Score $\neq$ select still applies: we are fitting the **probability model for draft**, not proving the NBA uses our score equation.

Maximum likelihood in one paragraph (no jargon yet)

For every player-season we see a **yes/no** outcome: drafted ($Y = 1$) or not ($Y = 0$).

We write down a **model** that assigns each player a probability p of being drafted, where p depends on ability A , congestion L^C , and the knobs λ and t .

Maximum likelihood means: choose λ and t so that, if the model were true, the **whole history of yes/no outcomes we actually saw** would be as **likely as possible**.

No new data. No new experiment. One pass over the historical panel: “Which parameters make our story hardest to reject?”

That is standard statistics — not a sim-only trick.

Step 1 — What goes into each row

One **row** = one player in one season.

Column	Meaning
A	Ability (PPM z-score within that season)
L^C	Leave-one-out congestion on his roster (depends on γ)
Y	1 if ever drafted from college, 0 if not
Roster	Fixed from real data (not re-simulated)

We are **not** moving players between teams in this fit. p already did its job when the panel was built.

Step 2 — From ability and congestion to a draft probability

This is the heart of the model. Read slowly; `MLE_basics.md` shows the algebra.

2a. A “draft appeal” number for each player

Before probabilities, each player i in season s gets a **single number** that says how attractive he looks **relative to everyone else that season**:

$$\text{appeal}_i = \frac{A_i}{t} - \lambda L_i^C$$

Plain English:

- **Higher ability $A \rightarrow$ higher appeal (good).**
- **Higher congestion $L^C \rightarrow$ lower appeal (crowded roster hurts), scaled by λ .**
- **Temperature t divides ability: smaller t means ability differences matter **more** (sharper ranking); larger t flattens everyone toward similar appeal.**

λ and t are what we **estimate**. A and L^C are **inputs** from the panel (L^C changes when γ changes — that is why we later “profile” γ).

2b. Turn appeal numbers into probabilities (within one season)

Draft probabilities must be **between 0 and 1** and **add up to 1** across all players **in that season** (we treat each season as its own pool). The standard way to do that is the **softmax** (explained step-by-step in MLE_basics):

$$p_i = \frac{\exp(\text{appeal}_i)}{\sum_{j \text{ in same season}} \exp(\text{appeal}_j)}$$

So: exponentiate each appeal, then divide by the season total. Highest appeal $\rightarrow$ highest p , but everyone shares one “pie” that sums to 1. **Important:** Softmax probabilities sum to **1** per season, not K . We get a **ranking**; K enters later in sanity checks and sim (**Step 5**), not inside this formula.

Step 3 — Bernoulli likelihood (yes/no trials)

Once each player has a model probability p_i , treat draft as a **coin flip** biased toward p_i :

- If $Y_i = 1$ (drafted): the model contributes a factor p_i — “how likely was a yes?”
- If $Y_i = 0$ (not drafted): the model contributes $(1 - p_i)$ — “how likely was a no?”

For one player:

$$P(Y_i | p_i) = p_i^{Y_i} (1 - p_i)^{1 - Y_i}$$

Multiply over **all** player-seasons (we assume independence). Take the **log** (products become sums — easier for the computer):

$$\ell = \sum_i \left[Y_i \log p_i + (1 - Y_i) \log(1 - p_i) \right]$$

Maximum likelihood = find λ and t (and optionally γ) that make **ℓ as large as possible**.

That sum ℓ is what the script prints as log-likelihood. Bigger (less negative) = better fit.

Step 4 — How the computer actually finds $\hat{\lambda}$ and $\hat{t}$

1. **Build the panel** — ability, L^C at $\gamma = 18$, draft labels.
2. **Coarse grid** — try many (λ, t) pairs on a grid (e.g. λ from 0.5 to 3, t from 0.01 to 10). Keep the pair with the best ℓ .
3. **Refine** — hand that pair to a numerical optimizer (L-BFGS-B: smooth hill-climbing with bounds) that adjusts λ and t until ℓ stops improving.
4. **γ profile (optional diagnostic)** — fix γ at 5, 8, 10, ..., 200; at **each** γ , rebuild L^C , repeat grid + refine for (λ, t) , record the best ℓ . Plot ℓ vs γ to see whether draft data pin γ down (they mostly do not).

Script: `sports/scripts/pd21_draft_bernoulli_mle.py`

Outputs: `../re_entry/HEROs_and_PASSes/pd21_mle/`

Step 5 — Where K fits (two phases — read this)

You are right to ask: **we do not pass a known K into the optimizer**. There is **no** second math step that “finalizes” $\hat{\lambda}$ and $\hat{t}$ using K . The story is two phases:

Phase A — Fit (what we already did)

- **Input:** every player’s Y (drafted or not). Each season has K_s drafted players — that is just **how many $Y = 1$** rows you see (e.g. 89 in one season).
- **Estimate:** λ, t by maximizing Bernoulli ℓ .
- **K is not in the formula.** Softmax makes $\sum p_i = 1$ per season, not K .
- **Output:** $\hat{\lambda} \approx 2.57$, $\hat{t} \approx 1.07$, and a p_i for every player. **Done.** Parameters do not get re-fit.

Phase B — Use known K (after the fit — no new λ, t search)

1. **Sanity check:** For each season, sort by p_i , take the top K_s (same K_s as counted from data), compare to real draftees. Mean overlap ≈ 12 picks — **check only**, not re-estimation.
2. **We do not** rescale p_i so they sum to K (not in v1).
3. **Sim / generative SELECT:** Use $\hat{\lambda}, \hat{t}$ to build weights, then draw **exactly K** draftees without replacement (K -draw). That is how the wind tunnel picks fake draft classes — **same parameters**, different selection rule than Bernoulli MLE.

If you wanted K inside the fit from the start: that would be a **K -draw likelihood** (different model) — not what we ran. See `MLE_basics.md` Part 5 and Part 10.

Hand example: `MLE_basics.md` Part 6 — five players, one season, $K = 2$, compute p and ℓ by hand.

What we found (2013–2021)

Quantity	Value	Read it as
λ^*	≈ 2.57	Congestion in the score helps explain who got drafted ($\lambda > 0$). Ability-only would miss something.
t^*	≈ 1.07	Moderate sharpness — not “only #1 matters” ($t \rightarrow 0$), not “everyone equal” ($t \rightarrow \infty$).
γ	18 fixed	Same as sim default. Profile shows ℓ nearly flat from $\gamma \approx 18$ to 100 — draft labels do not tightly identify γ .
Best ℓ	≈ -6919.1	Best score under $\gamma = 18$ (only meaningful relative to other λ , t , γ tries).

γ profile takeaway: ℓ is slightly best around $\gamma = 100$ on the grid, but λ^* and t^* barely move. We report **$\gamma = 18$** to stay aligned with the generative sim, not because MLE screams “18.”

Sanity (not headline): mean overlap between model top-K and real draftees ≈ 12 picks per season; recall on drafted players $\approx 12\%$. Soft probabilities $\neq$ perfect ranker — expected.

Two different uses of “temperature t ” (do not mix them up)

A. PD20 generative sweep (before MLE)

Question: In the **simulator**, if SELECT is soft (Gibbs) instead of hard top-K, does the inverted-U in draft rate vs teammate quality **survive**?

Method: Grid over t on fake leagues — **not** fitting real draft labels.

Answer: Yes, mostly $\rightarrow$ we were allowed to move on to empirical MLE.

Plot: `../re_entry/HEROs_and_PASSes/pd20_temperature/GRANDCHILD_temperature_select_sweep_2013_2021.png`

B. Empirical MLE (this fit)

Question: What t (and λ) best explain **real** draft yes/no outcomes?

Answer: $t^* \approx 1.07$.

Use **t^* from MLE** when you talk about the empirical fit. The PD20 sweep is the **gate** that justified soft SELECT in sim — not a substitute for t^* .

Why Bernoulli here, not “draw exactly K winners”?

The **simulator** often picks **K** draftees without replacement (weighted draws). You could write a likelihood that matches that exactly. For **empirical** draft MLE we use **independent Bernoulli trials** with softmax **p_i** instead (PD21 lock). Reasons in plain language:

1. **Tractable** — per-player factors multiply cleanly; optimizer is standard.
2. **Alex’s framing** — fit the **distribution of draft probability**, so high- **p** players match drafted players; K is in the **data**, not forced into $\sum p_i = K$.
3. **Sim can stay on K-draws** for pictures; at NCAA scale the gap is expected to be small.

See `MLE_basics.md` for the honest caveats (probabilities sum to 1 per season, not K; independence vs “exactly K picks”).

Hero (Layer A) — separate thread

The **ventile chart** (draft rate vs teammate quality) is **not** this fit.

- POST-QC: middle rise, **flat** elite tail.
- July inverted-U tail = pre-QC sensitivity only.

This document is about **λ , t , γ on draft labels**. Hero shape is documented elsewhere.

Questions that come up while reading

Is this really maximum likelihood?

Yes. We maximize the sum of log Bernoulli probabilities with **$p_i(\lambda, t, \gamma)$** from the softmax story above. JSON: `PD21_draft_bernoulli_mle_2013_2021.json`.

Why not fit γ in the main run?

Draft yes/no does not move γ much once λ and t adjust. We anchor $\gamma = 18$ for one line with the sim; joint (λ, γ, t) MLE is in the script if needed later.

Where is K? Are we fitting with a known K?

No — K is not in the likelihood. We fit λ and t using every row’s yes/no **Y**. Each season’s **K_s** is implicit (count of yeses). After the fit, we **check** top- **K_s** overlap and use **K-draw** in sim — but we do **not** run a second optimization to “apply K” to finalize λ^* , t^* . See **Step 5** and MLE_basics Part 5–6.

Where is p ?

Fitted separately on roster sorting (**H_sort**). Draft MLE uses **fixed** empirical rosters.

Files

Item	Path
This doc (overview)	3-Master_Plan/MLE/MLE_fit_explainer.md
Derivations & caveats	3-Master_Plan/MLE/MLE_basics.md
Slide deck	../re_entry/HEROs_and_PASSes/slides/auto/CHAR_PD21_MLE_fit_AUTO.pptx
Fit JSON	../re_entry/HEROs_and_PASSes/pd21_mle/PD21_draft_bernoulli_mle_2013_2021.json
γ profile plot	../re_entry/HEROs_and_PASSes/pd21_mle/PD21_draft_bernoulli_mle_2013_2021_gamma_profile.png
Regenerate slides + figure	python sports/scripts/build_pd21_mle_fit_slides.py --season-min 2013 --season-max 2021 --slides-only --force-figures

One-sentence summary

We assigned each player a draft probability from ability and congestion (knobs λ , $\mathbf{t}$), scored how well those probabilities explain every real drafted / not-drafted outcome, and picked $\hat{\lambda} \approx 2.6$, $\hat{\mathbf{t}} \approx 1.1$ with $\mathbf{y} = 18$ fixed — congestion in the score matters; the hero ventile chart is a different, parked thread.